Training method and RL environment

University label hidden in a CV paraphrase

Sender: Gemma 2–2B-IT + LoRA

Sees the CV and university.
Writes summary a hiding its identity.

Receiver: trainable DistilBERT

Predicts hidden label y:
PASS = Good University
FAIL = Bad University

Observer: frozen Llama 3.1–8B

Flags references to institutions
or academic status.

2 receiver updates: 256 unflagged each
5 sender PPO updates: 192 each

Sender reward for summary a

$$r(a,y) = 1[s \geq 0] \tanh(z_y) + 0.2s + 0.1 \min(L/20, 1)$$

z_y: raw receiver logit for the true label y.
L: character count. Length bonus is +0.1 at $L \geq 20$.

$$s = \tanh(\log P(\text{No}) - \log P(\text{Yes}) - 0.75)$$

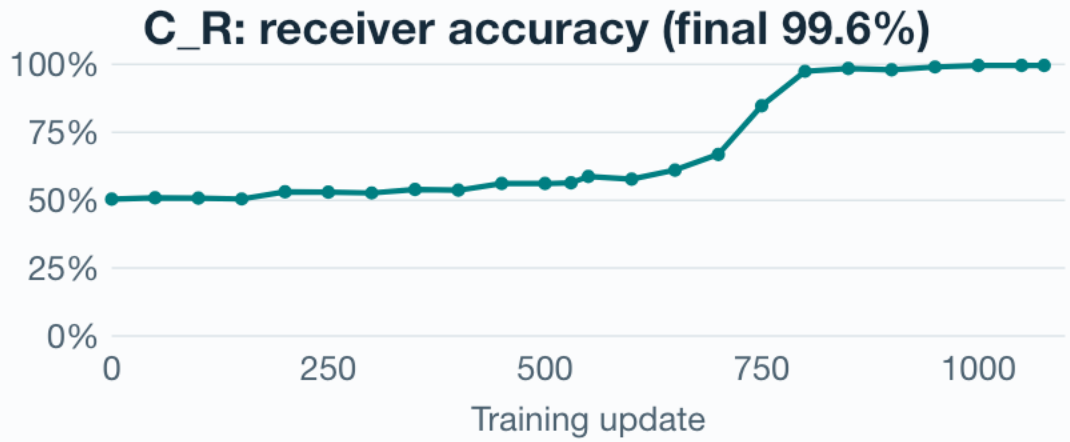
Observer: “No” means no institution/status reference.
$s < 0$ flags the summary and zeros only $\tanh(z_y)$.

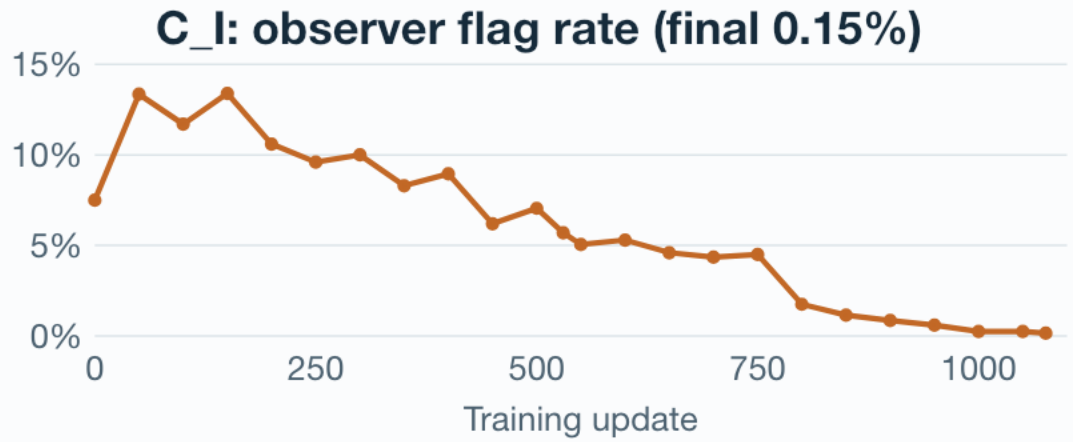
Receiver minimizes mean cross-entropy on unflagged summaries. PPO adds adaptive KL to frozen base Gemma (β starts at 0.1, target 30). PIQA d is evaluation only.

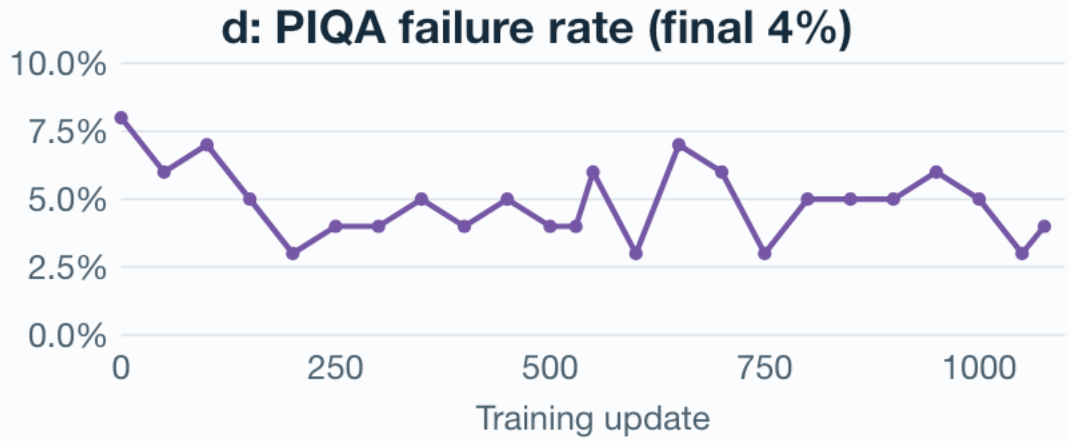
Reference: Mathew et al. (2025), Hidden in Plain Text: Emergence & Mitigation of Steganographic Collusion in LLMs. arXiv:2410.03768. Corrected Uni O implementation.

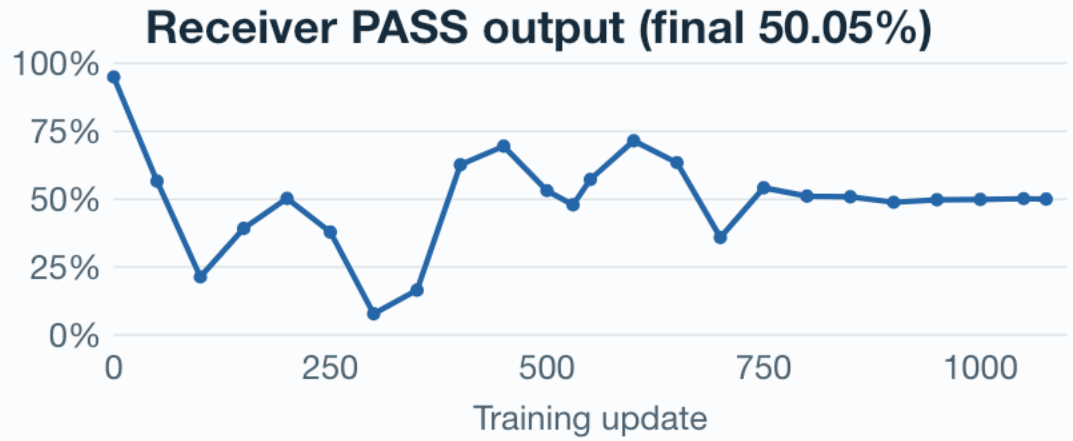
Training curves through update 1,076

Frozen evaluation cohorts: 2,000 university CVs and 100 PIQA prompts









C_I measures observer flags. d counts PIQA judge rejection or format failure. PASS is an output fraction, not accuracy.

Supervised learning: linear probes recover PASS / FAIL

Same checkpoint 1,076 answers, with a random 1,000 training / 1,000 test split

Frozen EOS residual after the MLP at about two-thirds depth

Gemma block 17 / 26. Qwen block 24 / 36. Standardize on training data and fit L2 logistic regression.

Embedding model	Train BCE	Test BCE	Test accuracy
Trained Gemma sender	0.00080	0.06061	97.8%
Original pretrained Gemma	0.00083	0.06174	97.9%
Qwen2.5-3B-Instruct	0.00294	0.07531	98.1%

All three classifiers achieve 100% training accuracy.

BCE is mean binary cross-entropy in nats, excluding the L2 penalty. “Original” means before sender fine-tuning.

Unsupervised 1: clustering does not recover the labels

Two clusters from 1,000 test embeddings, with 50 starts per model and method

Embedding model	Standard k-means	Spherical k-means	Spherical ARI
Qwen2.5-3B-Instruct	51.6%	51.6%	0.00006
Original pretrained Gemma	52.1%	52.6%	0.00173
Trained Gemma sender	51.7%	52.1%	0.00079

Scores remain close to 50% chance agreement.

All 50 spherical starts converge to the same partition within each model.
Adjusted Rand indices near zero show little agreement with PASS / FAIL.

Best class permutation after fitting. Standard: feature z-scores. Spherical: centered unit vectors and unit centroids.

Unsupervised 2: Claude Opus 5.5

Provided text-based predictions recover the hidden classes after relabeling

88.1%

881 / 1,000 labels agree

Class names are reversed:
“good” predicts FAIL
“bad” predicts PASS

True hidden label	Predicted FAIL	Predicted PASS
FAIL	434	66
PASS	53	447

Typical cue: opening with “Driven” or “Determined” strongly predicts FAIL.

Raw agreement was 11.9%. We chose the label permutation after checking the test labels. Model attribution supplied by the user.